Measuring Explainability and Trustworthiness of Power Quality Disturbances Classifiers Using XAI—Explainable Artificial Intelligence

Ram Machlev, *Student Member, IEEE*, Michael Perl, Juri Belikov, *Senior Member, IEEE*, Kfir Yehuda Levy, and Yoash Levron, *Senior Member, IEEE*

***Abstract*—Advanced machine learning techniques have recently demonstrated outstanding performance when applied to power quality disturbance (PQD) classification. Nevertheless, a possible problem is that power experts may find it hard to trust the results of such algorithms, if they do not fully understand the reasons for their outputs. In this light, this article suggests a method that explains the outputs of PQD classifiers, using explainable artificial intelligence (XAI). The method operates as follows: first, various XAI techniques and classifiers are combined and scored based on their explanations during the validation step. Then, the best combination of classifier and XAI technique for each disturbance is used on the testing set, such that the classifier outputs are more transparent. To accomplish these steps, a definition for the correct explanation in PQD is given. Also, to determine the quality of an explanation for a certain output, we propose an evaluation process to measure the *explainability score* for each XAI technique and classifier. By means of this approach, the PQD classifier outputs are optimized to be both accurate and easy to understand, allowing experts to make informed and trustworthy decisions.**

***Index Terms*—Convolutional neural network (CNN), deep-learning (DL), energy, evaluation metrics, explainable artificial intelligence (XAI), power quality disturbances (PQDs), power.**

Manuscript received June 17, 2021; revised September 12, 2021; accepted November 3, 2021. Date of publication November 9, 2021; date of current version May 6, 2022. This work was supported in part by the Zuckerman Fund for Interdisciplinary Research in Machine Learning and Artificial Intelligence at the Technion, in part by the Technion Center for Machine Learning and Intelligent Systems (MLIS), and in part by the Nancy and Stephen Grand Technion Energy Program (GTEP), in association with the Guy Sella Memorial Project. The work of Yoash Levron was supported by Israel Science Foundation, under Grant 1227/18. The work of Kfir Yehuda Levy was supported by Israel Science Foundation, under Grant 447/20. Paper no. TII-21-2497. *(Corresponding author: Ram Machlev.)*

Ram Machlev, Michael Perl, Kfir Yehuda Levy, and Yoash Levron are with the Andrew and Erna Viterbi Faculty of Electrical and Computer Engineering, Technion—Israel Institute of Technology, Haifa 3200003, Israel (e-mail: ramm@campus.technion.ac.il; michael.perl@campus.technion.ac.il; kfirylevy@technion.ac.il; yoashl@ee.technion.ac.il).

Juri Belikov is with the Department of Software Science, Tallinn University of Technology, 12618 Tallinn, Estonia (e-mail: juri.belikov@taltech.ee).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TII.2021.3126111.

Digital Object Identifier 10.1109/TII.2021.3126111

I. Introduction

With the evolution of deep learning (DL), better classifiers and machine learning (ML) algorithms are being developed for power system applications [1], [2]. However, despite the evident success of such algorithms, an inherent difficulty is that since ML algorithms are often very complex, it may not be clear how or why they make certain decisions. Therefore, experts in the power system field may find it hard to trust the decisions and recommendations made by ML-based algorithms, limiting their practical usefulness. This difficulty is especially prominent in cases that require high level of reliability, which are common in the energy industry.

Due to this challenge, in the last couple of years new techniques and principles are being developed to improve the *explainability* of ML models, such that their output can be better understood. This concept is known in the literature as explainable artificial intelligence (XAI) [3]. The goal of XAI is to help researchers, developers, domain experts, and users to better understand the inner operation of ML models, while preserving their high performance and accuracy [4], [5]. The use of XAI for power and energy applications is still new, and only a few studies have been published so far. Two examples are [6] and [7], which suggest explainable forecasting methods for cooling and solar PV related applications.

Recently, several works have implemented DL techniques for classification of power quality disturbances (PQD), which in most cases outperform traditional classification algorithms. One of the first works that uses deep learning for PQD classification is [8], which proposes various architectures such as convolutional neural network (CNN) and long short-term memory. Other implementations focus on specific architectures such as multilabel classification [9], multitask learning deconvolution networks [10], multifusion CNNs [11], CNNs with inception modules [12], and residual networks [13]. In addition, several of these studies use a preprocessing stage that transforms the original power signal, employing, for example, the curvelet transform [14], compressed sensing techniques [15], or Wigner distribution functions [16]. All of the above deep learning techniques seem to have advantages compared to traditional algorithms in terms of efficiency, noise immunity, and accuracy, which may be higher than 99% as shown in [17]. However, a core obstacle, as mentioned above, is that regardless of their

1551-3203 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

widespread adoption and high accuracy in PQD classification tasks, predictions made by DL are hard to explain. Hence, power experts may find it hard to trust the decisions made by such algorithms if they do not clearly understand why they reach a certain classification, and how they treat real-world data.

Considering this gap, the main goal of this article is to present a method that explains the results of PQD classifiers, using XAI. The method operates in two stages: in the first stage, various XAI techniques and classifiers are combined and scored based on their explanations during the validation step. Then, in the second stage, the best combination of classifier and XAI technique for each disturbance is chosen and used on the testing set, such that the classifier decisions are more transparent. In general, one of the main challenges in XAI is the unclear definition of a correct explanation [18]. Therefore, one of the contributions of this article is to provide a clear definition of a correct explanation for PQD applications. This definition, named here the "ground-truth" explanation, allows a precise evaluation of how much a given explanation matches the correct explanation. Furthermore, another challenge is how to measure and evaluate the explanation for PQD classifiers. Accordingly, another contribution of this article is to develop an evaluation process to measure an *explainability score* for each XAI technique and classifier. In this process two evaluation metrics are suggested, the first metric is based on a binary score, and the second metric is based on the intersection over union (IoU) score. Note that this process may be adapted for other power system applications in which the correct explanation can be clearly defined. These ideas are demonstrated using four different CNN-PQD classifiers and three XAI techniques: occlusion-sensitivity [19], locally interpretable model-agnostic explanations (LIME) [20], and gradient class activation mapping (Grad-CAM) [21]. To conclude, the contributions of this article are the following ones.

1) A new method that explains the results of PQD classifiers, based on recently developed XAI techniques. By implementing this method, the PQD classifier outputs are optimized to be both accurate and easy to understand. The method operates as follows: first, various XAI techniques and classifiers are combined and scored based on their explanations during the validation step. Then, the best combination of classifier and XAI technique for each disturbance is used on the testing set, such that the classifier outputs are more transparent in comparison to existing techniques. In addition, this article is the first to use XAI for PQD classification task.
2) A precise definition for a correct explanation in the context of PQD applications is given. To the best of our knowledge, in most applications (including in the energy and power domain) there is no clear definition of a correct explanation. Thus, it is a challenge to determine whether an explanation of a decision made by a classifier is legitimate. This clear definition allows a comparison between an output of a XAI technique and the correct explanation. By using this definition, the implementation and integration of XAI for PQD in the industry will be simpler and smoother.
3) An evaluation process which allows power experts and researchers to measure how good and robust a PQD classifier is by means of trustworthiness and interpretability. Many PQD classifiers have high accuracy but their explainability is not considered, let alone evaluated. This process is achieved by measuring the explainability score of a given PQD classifier and XAI technique.

The rest of this article is organized as follows. Section II provides background on PQD, the suggested DL classifiers, and XAI techniques. Section III presents the definition of explanation, the explainablity evaluation process, and the optimal explanation method for PQD. Section IV shows experimental results. Finally, Section V concludes this article.

II. DL Classifiers and XAI Techniques for PQD

A. Overview and Modeling PQDs

Power quality (PQ) is a measure of the degree to which voltage and current waveforms comply with established specifications [22]. In the last two decades, PQ monitoring tools are becoming a necessity [23]. One reason for the need of such tools is the continuing integration of nonlinear generators and loads in power grids, most of them based on power electronics technology, which may inject high-order voltage and current harmonics [24]. Thus, many recent studies focus on detection and classification of PQDs [25]–[28].

The type of disturbances that are used in this article, as well as their characteristic mathematical model and typical parameters are given in [29, Table I]. The parameter limits are in accordance with the IEEE-1159 standard [30]. In this table, nine disturbances are simple disturbances while six are complex disturbances, which are composed of two or more simple PQ disturbances.

B. DL Classifiers for PQD

The first network, which is taken from [17], consists of rectified linear units (ReLU)s, max-pooling layers, and batch-normalization layers. For all convolutional and max-pooling layers the kernel size is 3, and the stride value is 1, and its architecture is shown in Table I. The second network is based on [11], but without the fusion layers, and its architecture is shown in Table II. In the second CNN, the kernel size is 13 for all convolutional and max-pooling layers. The third network is based on [13], and uses residual layers as shown in Fig. 1(a). The architecture of this network is presented in Table III. In the third CNN, for all convolutional layers the kernel size is 16 and the stride value is 1 and for all max-pooling layers the kernel size is 2 and the stride value is 2. The fourth network is based on [12], which uses inception-residual blocks as shown in Fig. 1(b). The architecture of this network is presented in Table IV.

C. Suggested XAI Techniques

In [5], the term *explanation* is defined as follows: additional information, generated by an external algorithm (or by the classifier itself) that describes the relevant features of an input instance for a particular output classification. Mathematically,

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CNN-1 ARCHITECTURE FROM [17]

#	Layer	Parameters	Activation
1	Convolution (1x3)	Filter size = 32, Stride = 1	ReLU
2	Convolution (1x3)	Filter size = 32, Stride = 1	ReLU
3	MaxPooling (1x3) , Stride = 1		
4	Batch normalization		
5	Convolution (1x3)	Filter size = 64, Stride = 1	ReLU
6	Convolution (1x3)	Filter size = 64, Stride = 1	ReLU
7	MaxPooling (1x3) , Stride = 1		
8	Batch normalization		
9	Convolution (1x3)	Filter size = 128, Stride = 1	ReLU
10	Convolution (1x3)	Filter size = 128, Stride = 1	ReLU
11	Global MaxPooling		
12	Batch normalization		
13	Fully connected	Size = 256	ReLU
14	Fully connected	Size = 128	ReLU
15	Batch normalization		
16	Fully connected	Size = 16	Softmax

TABLE II
CNN-2 ARCHITECTURE BASED ON [11]

#	Layer	Parameters	Activation
1	Convolution (1x13)	Filter size = 8, Stride = 1	ReLU
2	MaxPooling (1x13)	Stride = 2	–
3	Batch normalization		
4	Convolution (1x13)	Filter size = 8, Stride = 1	ReLU
5	MaxPooling (1x13)	Stride = 18	–
6	Batch normalization		
7	Convolution (1x13)	Filter size = 8, Stride = 1	ReLU
8	MaxPooling (1x13)	Stride = 2	–
9	Batch normalization		
10	Fully connected	Size = 300	ReLU
11	Batch normalization		
12	Fully connected	Size = 16	Softmax

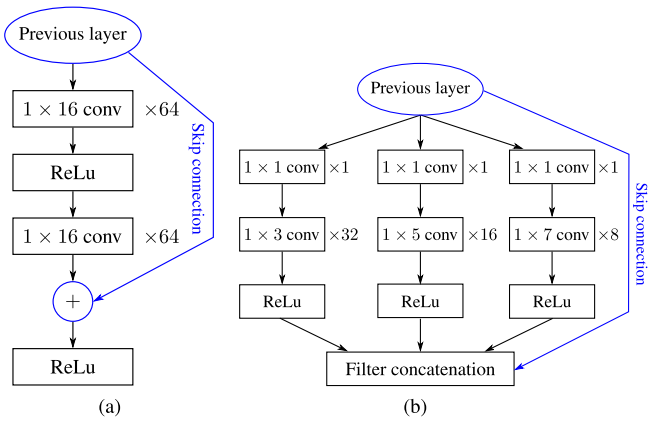


Fig. 1. Special convolution layers that were used in this article based on [13] and [12]. (a) Residual block used for CNN-3. (b) Inception-Residual block used for CNN-4.

TABLE III
CNN-3 ARCHITECTURE [13] WITH RESIDUAL BLOCK FROM FIG. 1(A)

#	Layer	Parameters	Activation
1	Convolution (1x16)	Filter size = 64, Stride = 1	ReLU
2	MaxPooling (1x2)	Stride = 2	–
3	Residual block		
4	Residual block		
5	MaxPooling (1x2)	Stride = 2	–
6	Residual block		
7	Residual block		
8	MaxPooling (1x2)	Stride = 2	–
9	Fully connected	Size = 256	ReLU
10	Fully connected	Size = 16	Softmax

TABLE IV
CNN-4 ARCHITECTURE [12] WITH INCEPTION-RESIDUAL BLOCK FROM FIG. 1(B)

#	Layer	Parameters	Activation
1	Inception-Residual block		
2	Inception-Residual block		
3	Inception-Residual block		
4	Inception-Residual block		
5	Inception-Residual block		
6	fully connected	size = 100	dropout = 0.8
7	fully connected	size = 16	Softmax

for a ML model f with input $x \in \mathbb{R}^z$ and output prediction of $\bar{y}$ for class c, an explanation map $g \in \mathbb{R}^z$ is generated to describe the feature importance, or the relevance of that input to the class output. In this article $g = G(x, f, \bar{y})$ is a heatmapping function such that the value for each variable $g[n]$ indicates how important the feature $x[n]$ in the model f for representing the target class c. There are various XAI techniques and approaches [18], most of them are dedicated to DL models. Here, three popular XAI techniques that are used in this article are presented.

1) Occlusion-Sensitivity: Occlusion-sensitivity was introduced in [19], and is used for understanding the prediction of a CNN based on the input image features. The sensitivity of the trained CNN is estimated by iteratively occluding a different location of the input using a small mask. The explanation output is a heatmap, which explains what changes in the input affect the classification output, by highlighting it with colors that indicate the importance of each pixel for the classification.

2) Grad-Cam: Grad-CAM is a method dedicated to CNNs, which uses gradients to understand how features after the final convolution layer affect the classification [21]. To obtain the explanation map for class c, $g^c_{\text{Grad-CAM}} \in \mathbb{R}^{W \times H}$, the gradients are first set to zero for all classes except for the desired class. Then, the gradients of the classification score y^c (before the final layer) with respect to K feature map activations $\{A^k\}_{k \in \{1,\ldots,K\}}$ of a convolutional layer are computed, i.e., $\partial y^c / \partial A^k$. Then, the gradients are globally averaged over the width and height

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

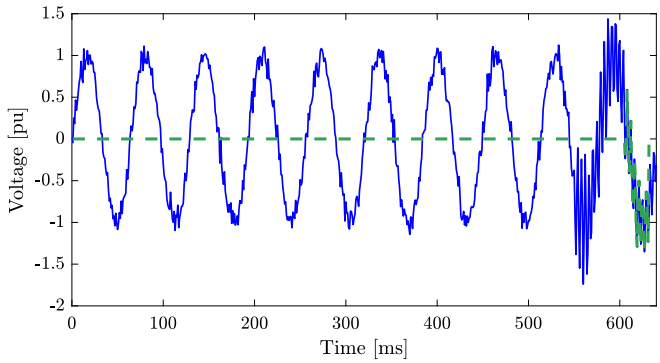


Fig. 2. XAI for PQD classification. Example 1: explanation of CNN-4 prediction of oscillatory transient signal using LIME.

dimensions (w and h, respectively) to obtain the neuron importance weights α_k^c [21, (1)]:

$$\alpha_k^c = \frac{1}{WH} \sum_w \sum_h \frac{\partial y^c}{\partial A_{w,h}^k}. \tag{1}$$

The weight α_k^c represents a partial linearization of the deep network downstream from A, and captures the "importance" of feature map k for a target class c. Then, a weighted combination of activation maps is input to a ReLU to obtain [21, (2)]

$$g_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_{k=1}^{K} \alpha_k^c A^k\right). \tag{2}$$

3) LIME: Work [20] introduced LIME, which attempts to find the importance of contiguous features in an input toward the output class. It does this by finding a binary vector, which represents the presence of continuous features that provide the reasons for the predicted output. LIME can be used with any ML model, unlike occlusion-sensitivity and Grad-CAM, which are suggested mainly for DL classifiers. Formally, consider $\xi \in \Xi$, a model from a class of potentially interpretable models Ξ. The model ξ can be decision trees, linear models, or other models of varying interpretability. Let $\Omega(\xi)$ measure explanation complexity, $\pi_x(z)$ be a proximity measure between two input instances x and z. The function $L(f, \xi, \pi_x)$ represents the unfaithfulness of ξ in approximating a ML model f in the locality defined by π_x. Then, explanation g for the input data sample x and ML model f is given by the LIME equation [20, (1)]:

$$g(x) = \arg\min_{\xi} \left(L(f, \xi, \pi_x) + \Omega(\xi)\right). \tag{3}$$

D. Motivation for XAI in PQD Classifiers: Examples

Decisions related to PQ require a high level of accountability and transparency, since they may effect the reliability and quality of the entire power system. An example of an oscillatory transient disturbance which is classified using trained CNN-4 and explained by LIME is presented in Fig. 2. In this example, the main features that explain the prediction are colored in green. It can be seen that these main features are indeed part of the oscillatory transient; however, it does not include the whole disturbance. Another example is presented in Fig. 3. In this example, a sag signal is explained by occlusion-sensitivity on the four suggested DL classifiers, which all classified the input as "Sag." Nonetheless, the sensitivity map is different for each classifier. The main features that positively contribute to the explanation for the first and second networks are only part of the disturbance, while the explanation for the third network is related, almost perfectly, to the entire disturbance. Furthermore, the fourth network decision is explained based on irrelevant data since the main features for the decision are not related to the Sag. To conclude, based on the suggested examples, there is a correlation between the correct prediction and the explanation. Notwithstanding, it is still not clear how to evaluate the explanation of a classifier and XAI technique.

III. Proposed Explanation Method

A. Definition of a Correct Explanation in PQD

Unlike many other classification tasks, the correct explanation for PQD can be defined unequivocally. Specifically, it can be specified as the time in which the disturbance appears, and can be present as a binary vector that represents the presence or absence of the disturbance at each sample time. Consider $x_0 = \sin[\omega n]$ as a "normal" discrete signal, which represents a normalized voltage waveform, and let x_d be a discrete signal with a single disturbance. Based on this, below we formulate the natural definition for a correct explanation of a single PQD.

Definition 1 (Single PQD Explanation): A vector v_e, defined as

$$v_e[n] := \begin{cases} 1, & \text{if } x_0[n] \neq x_d[n], \\ 0, & \text{otherwise} \end{cases} \quad \forall n \in \{1, \ldots, N\} \tag{4}$$

is called the ground-truth explanation of x_d.

Now, consider X_{dc} as a discrete signal with a complex disturbance, which consists of m disturbances $x_{d_1}, x_{d_2}, \ldots, x_{d_m}$ each with a ground-truth explanation vector $v_{e_1}, v_{e_2}, \ldots, v_{e_m}$. Then, the definition for the ground-truth explanation of complex PQD can be formulated as follows.

Definition 2 (Complex PQD Explanation): A vector V_e, defined as

$$V_e[n] := \begin{cases} 1, & \text{if } x_0[n] \neq x_{d_i}[n], \\ 0, & \text{otherwise} \end{cases} \quad \forall i$$

$$= \bigwedge_{i=1}^{m} v_{e_i}[n] \qquad \forall n \in \{1, \ldots, N\} \tag{5}$$

is called the ground-truth explanation of X_{dc}, where $\bigwedge$ denotes the AND operator.

Note that for $m = 1$ the definition of the *complex* PQD explanation reduces to its *single* version.

B. XAI Evaluation Process: Explainability Score

Here, an evaluation process to measure the explainability for a single input with any classifier and any XAI technique is provided. This process is shown in Fig. 4 and described in Algorithm 1.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

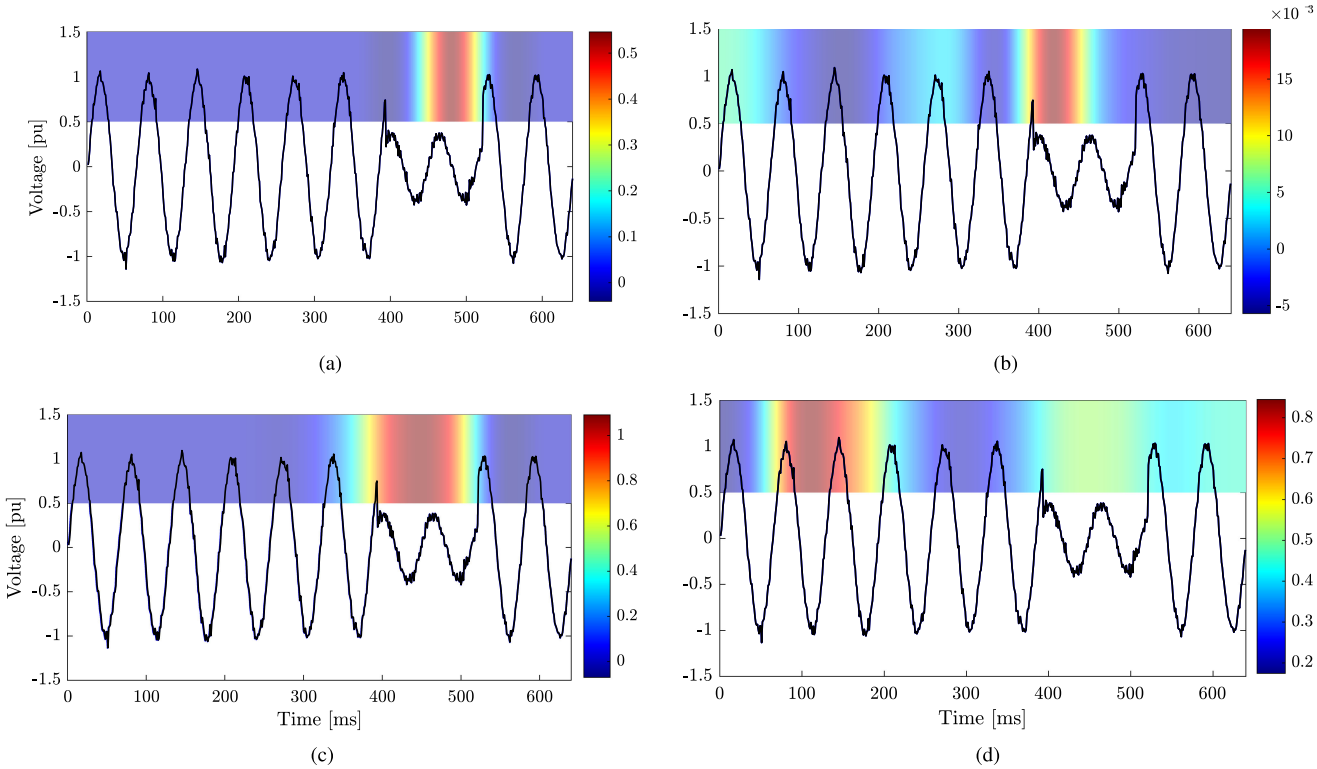


Fig. 3. XAI for PQD classification. Example 2: explanation of Sag signal using occlusion-sensitivity on four different DL classifiers. All networks predict the signal as "Sag."

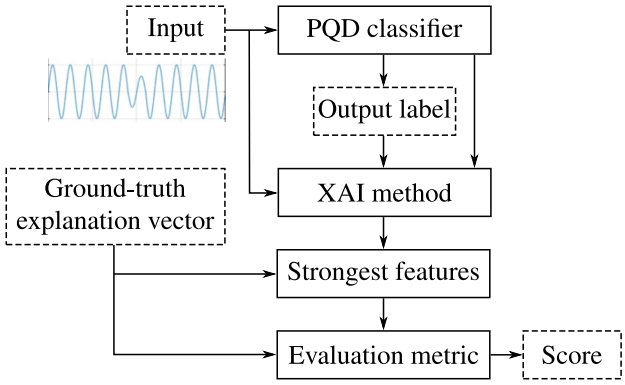


Fig. 4. Evaluation process to measure the explainability score for a single input with any classifier and any XAI technique.

Algorithm 1: ExplainScore.

Inputs:
x: signal input,
V_e: ground-truth explanation vector,
$\text{CNN}_{\text{trained}}$: trained classifier,
$\text{XAI}_{\text{technique}}$: used XAI technique.

begin
(1) $\hat{Y} \leftarrow \text{CNN}_{\text{trained}}(x)$;
(2) $H_{\text{map}} \leftarrow \text{XAI}_{\text{technique}}(x, \text{CNN}_{\text{trained}}, \hat{Y})$;
(3) $L \leftarrow$ number of nonzero elements in V_e;
(4) $\hat{V}_e \leftarrow \text{StrongFeatures}(H_{\text{map}}, L)$ ▷ using Algorithm 2;
(5) *Score* $\leftarrow \text{EvaluationMetric}(\hat{V}_e, V_e)$;
end
Output: *Score*: score for explainability of input x when using $\text{CNN}_{\text{trained}}$ classifier and $\text{XAI}_{\text{technique}}$.

In the algorithm $\text{StrongFeatures}(H_{\text{map}}, L)$, the strongest features in the output H_{map} of the XAI technique are found, where the number of features L is the number of nonzero elements in the ground-truth explanation vector V_e. This functionality is described in Algorithm 2 and its output is the estimated explanation vector $\hat{V}_e$. The command $\text{EvaluationMetric}(\hat{V}_e, V_e)$ evaluates the score of the estimated explanation vector in comparison to the ground-truth explanation vector. This explainability score can be measured using two evaluation metrics for each XAI technique on each classifier. The first metric is based on a binary score and the second metric is based on the IoU score. The definitions of these metrics are presented next.

Definition 3 (Hard Explainability Score): The *score* of the hard explainability is 1 if and only if $\exists n \in \{1, \ldots, N\}$ such that $(\hat{V}_e[n] = 1 \wedge V_e[n] = 1)$, otherwise the *score* of the hard explanation is 0.

For the example in Fig. 3, the hard explainability *score* is 1 for all networks except of network 4 with the *score* 0.

Definition 4 (Soft Explainability Score): The *score* of the soft explanation is defined as [31, (1)]

$$\text{Score} := \frac{\text{Area of Overlap}(\hat{V}_e, V_e)}{\text{Area of Union}(\hat{V}_e, V_e)} = \frac{\hat{V}_e \cap V_e}{\hat{V}_e \cup V_e}. \quad (6)$$

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2: StrongFeatures.

Inputs:
H_{map}: real-valued vector of feature contribution to explanation,
L: number of nonzero elements in V_e.

begin
  $\text{Ind}[1,\ldots,L] \leftarrow$ indexes of the L highest values in H_{map};
  Mark all elements of $\hat{V}_e$ as 0;
  Mark the estimated explanation $\hat{V}_e[\text{Ind}] = 1$;
end
Output: $\hat{V}_e$ estimated explanation vector.

This score is a number (from 0 to 1) that specifies the amount of overlap between the estimated explanation and the ground-truth explanation. For example, in Fig. 3, the soft explainability score *score* is 0.54 for CNN-1, 0.475 for CNN-2, 0.865 for CNN-3, and 0 for CNN-4. While the hard score does not fully utilize all the available information, the soft score uses all of the information from the strongest features. In the soft score, the reward is higher when the predicted strongest features heavily overlap with the ground-truth features.

It is important to mention that the use of ground-truth explanation as proposed in this article is not suggested in other studies.

C. Explainability Score for Multiple Classifiers and XAI Techniques

Assume there exists a set of trained classifiers $\{\text{CNN}^1_{\text{trained}},\ldots,\text{CNN}^C_{\text{trained}}\}$ and a set of XAI techniques $\{\text{XAI}^1_{\text{technique}},\ldots,\text{XAI}^\Psi_{\text{technique}}\}$. Then, the most suitable combination for explanation can be found for each disturbance. By using Algorithm 1 on the validation set, the average score for each disturbance on each combination of $\{\text{CNN}^c_{\text{trained}},\text{XAI}^\psi_{\text{technique}}\}$ can be calculated as shown in Algorithm 3, where $c \in \{1,\ldots,C\}$ and $\psi \in \{1,\ldots,\Psi\}$.

Based on the average scores of the validation set, the combinations with the highest score should be chosen for the inference stage (i.e., testing set) in order to present the best explanation for a specific disturbance input. A look-up-table (LUT) with D elements is used to store the combination of a classifier and XAI method with the highest average explainability score for each disturbance $d \in \{1,\ldots,D\}$.

IV. Performance Analysis

A. PQD Classifiers: Setup and Accuracy Results

All networks are configured based on the setups shown in Table V. The signals at each mini-batch are selected randomly for the back-propagation stage in the training process. The main training parameters in Table V are determined using parameter boundary values and trail-error (random) search. To verify the robustness of the classifiers and these parameters, the cross-validation method is used. The selected loss function is cross-entropy, and the optimizer is "Adam." This optimizer shows good performance in terms of computational efficiency,

Algorithm 3: ExplainScorePerDisturbnace.

Inputs:
X^{DW}: validation set which consists of D disturbances each with W signals,
V_e^{DW}: ground-truth explanation vectors for validation set,
Λ^C: set of trained classifiers $\{\text{CNN}^1,\ldots,\text{CNN}^C\}$,
χ^Ψ: set of XAI techniques $\{\text{XAI}^1,\ldots,\text{XAI}^\Psi\}$.

begin
  for $c \leftarrow 1$ *to* C **do**
    for $\psi \leftarrow 1$ *to* Ψ **do**
      for $d \leftarrow 1$ *to* D **do**
        $\text{ScoreAvg}[c][\psi][d] =$ ▷ using Algorithm 1
        $\frac{1}{W}\sum_{w=1}^{W} \text{ExplainScore}(X[d][w], V_e[d][w], \Lambda[c], \chi[\psi])$
      end
    end
  end
end
Output: $\text{ScoreAvg}\{C,\Psi,D\}$: average explainability score for any disturbance, for each classifier and each XAI technique.

TABLE V
TRAINING PARAMETERS

Parameter / CNN	1	2	3	4
Mini batch size	64		256	160
Max epochs	43	50	60	100
Loss function	Cross-entropy			
Optimizer	Adam			
$L2$ regularization	0.0001			
Initial learning rate (m_0)	0.01			
Learning drop factor (df)	0.5	0.2		
Learning drop period (dp)	10	5		

avoidance of local optimum, finding a high precision solution, convergence time, and reducing overfitting [32]. The learning rate is defined as [33, (3.52)]

$$\text{Learning Rate} = m_0 df^{\lfloor \frac{1+\text{epoch}}{dp} \rfloor} \tag{7}$$

such that it is initialized with m_0, and is dynamically reduced by a factor of df (learning drop factor) after each dp epochs (learning drop period) in case the loss function does not decrease. This dynamic learning rate strategy helps to avoid overfitting [33]. The training process is carrying out on NVIDIA GeForce RTX 2080 TI GPU with MATLAB DL toolbox.

The database is generated using a synthetic dataset generator that is designed to work with 16 types of PQDs, whose details are provided in [29]. The sampling rate for all signals is 3.2 kHz, the nominal frequency is set to 50 Hz ($\omega = 2 \cdot 50\pi$) and the duration is ten cycles. The generated dataset consists of 4800 signals of each disturbance type, for a total of 76 800 signals with noise which varies from 20 to 50 dB. The training set includes 80% of the samples from the complete dataset, the validation set includes 10% of the samples and the testing set includes the remaining 10%. These datasets are split randomly and each disturbance is represented by the same number of input samples, i.e., the datasets are balanced.

PQD classifiers are evaluated based on diverse metrics. Most of these metrics formulate as binary classification tasks, and

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

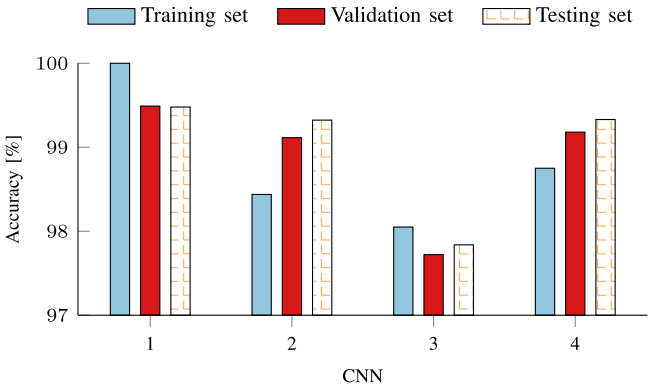


Fig. 5. Accuracy results for each classifier.

TABLE VI
SIMULATION AVERAGE EVALUATION METRICS RESULTS FOR THE TESTING SET PER CLASSIFIER

	CNN 1	CNN 2	CNN 3	CNN 4
Accuracy [in %]	99.479	99.323	97.838	99.322
F-measure [in %]	99.484	99.33	97.849	99.329

have three possible outcomes: true positive (TP) is the number of times a PQD is correctly detected, false positive (FP) is the number of times other disturbances are wrongly detected as a specific disturbance, and false negative (FN) is the number of times a specific disturbance is wrongly detected as other disturbances. *Accuracy* is a commonly used evaluation metric and is defined as [29, (1)]

$$\begin{aligned}\text{Accuracy} &= \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \\ &= \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \end{aligned} \tag{8}$$

In this article, we also use the F-measure metric, which is common in the information retrieval domain. The F-measure is the harmonic mean of the *precision* and *recall* measures, and is given by [34, (1)]

$$F\text{-measure} = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{9}$$

where

$$\text{precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{10}$$

The average classification accuracy of the training, validation, and testing sets is presented in Fig. 5. It can be seen that all CNN networks accuracy are higher than 97.5% and CNN-1 (from [17]) achieves the best accuracy. Also, Table VI shows the average Accuracy and F-measure for the testing set.

B. Explainability Score Process and Results

After training the classifiers, the average score per disturbance for each classifier and each XAI technique was calculated for the validation set based on Algorithm 3. Tables VII and VIII present the hard and soft explainability scores accordingly. The total average explainability score for each table is calculated as follows.

$$\begin{aligned}\text{Total Score} &= \frac{1}{DW} \\ &\times \sum_{d=1}^{D}\sum_{w=1}^{W} \text{ExplainScore}(X[d][w], V_e[d][w], \Lambda[c], \chi[\psi]) \end{aligned} \tag{11}$$

where all variables can be found in Algorithm 3 and ExplainScore is Algorithm 1.

For the hard explainability score, it can be seen that the best combination is the CNN-3 classifier with Grad-CAM technique. Furthermore, Grad-CAM gives the highest score for nine disturbances when it is used with CNN-3 and CNN-4. the score when using occlusion-sensitivity method is the highest for spike and notch disturbances for all networks. The LIME technique does not achieve a high score for any disturbance. Also, disturbances with harmonics and flicker get a score of 1 for all networks and XAI techniques, since there are many strong features due to the frequent appearance of the disturbance. For the soft explainability score, it can be seen that the best combination is the CNN-1 with occlusion-sensitivity method. Moreover, occlusion-sensitivity gives the highest score for nine disturbances when it is used with CNN-1 or CNN-3. When using Grad-CAM the score is the highest for five other disturbances, also, when combined with CNN-1 or CNN-3. The LIME technique achieved the highest score for flicker with CNN-3 classifier.

Based on the average scores results of the validation set, the combinations with the highest score are chosen for the inference stage (i.e., testing set) for each disturbance. This is done using a LUT with 15 elements, which is used to store the combination of a classifier and XAI method with the highest average explainability score for each disturbance $d \in \{1, \ldots, 15\}$. Tables IX and X present the LUT for the hard and soft scores accordingly. In these tables, the explainability score results for the testing set is presented as well. In the LUT for the hard explanation, CNN-3 is used four times while CNN-4 is used six times and Grad-CAM is the most used XAI technique. The total explainability score is 0.906, a higher score then any individual combination presented in Table VII. Similarly, in the LUT for the soft explanation, the total explainability score is 0.3235 a higher score then any individual combination presented in Table VIII. CNN-3 is used nine times while CNN-1 is used five times and occlusion-sensitivity is the most used XAI technique. Note that the best combinations for the hard score are not equal to the best combinations for the soft score. These are two different approaches for evaluation and the user should choose the score that is more relevant for his needs. For example, a hard score may be chosen if the user requires a minimum acceptable explanation, this may involve changing the threshold of pixels of overlap. The soft score may be chosen if the user values the highest precision explanations.

C. Real-Life Data Experiment

To evaluate the robustness of the proposed method an experiment using real-life electrical data was performed as well.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
AVERAGE HARD EXPLAINABILITY SCORE PER NETWORK/XAI/DISTURBANCE FOR THE VALIDATION SET

#	Disturbance	CNN-1	CNN-2	CNN-3	CNN-4
	XAI Techniques		{Occlusion-sensitivity / Grad-CAM / LIME}		
1	Sag	{0.694, 0.971, 0.331}	{0.613, 0.748, 0.206}	{0.889, 0.796, 0.68}	{0.818, **0.987**, 0.137}
2	Swell	{0.76, 0.971, 0.98}	{0.835, 0.75, 0.97}	{0.773, **0.989**, 0.769}	{0.846, 0.985, 0.93}
3	Interruption	{0.625, 0.244, 0.073}	{0.515, 0.646, 0.23}	{0.137, **0.95**, 0.0542}	{0.592, 0.812, 0.175}
4	Harmonics	{1, 1, 1}	{1, 1, 1}	{1, 1, 1}	{1, 1, 1}
5	Flicker	{1, 1, 0.99}	{1, 1, 0.99}	{1, 1, 1}	{1, 1, 1}
6	Oscillatory transient	{0.785, 0.82, 0.44}	{0.627, 0.465, 0.539}	{0.579, 0.973, 0.72}	{0.644, **0.996**, 0.26}
7	Impulsive transient	{0.021, 0.027, 0.039}	{0.025, 0.023, **0.048**}	{0.0396, 0.0354, 0.0417}	{0.0313, 0.135, 0.0271}
8	Notch (periodic)	{0.865, 0.83, 0.28}	{0.88, 0.873, 0.102}	{**0.958**, 0.758, 0.76}	{0.85, 0, 0.129}
9	Spike	{**0.94**, 0.77, 0.813}	{0.848, 0.69, 0.571}	{0.856, 0.427, 0.785}	{0.856, 0.535, 0.9}
10	Sag with harmonics	{0.827, 0.967, 0.696}	{0.794, 0.673, 0.498}	{0.846, 0.869, 0.227}	{0.735, **0.977**, 0.473}
11	Swell with harmonics	{0.94, 0.9, 0.67}	{0.852, 0.72, 0.896}	{0.898, 0.9, 0.975}	{0.881, **0.977**, 0.93}
12	Interruption with harmonics	{0.515, 0.3125, 0.0521}	{0.65, 0.71, 0.14}	{0.389, 0.948, 0.046}	{0.167, **0.96**, 0.063}
13	Flicker with harmonics	{1, 1, 1}	{1, 1, 1}	{1, 1, 1}	{1, 1, 1}
14	Flicker with sag	{0.66, 0.86, 0.56}	{0.383, 0.692, 0.127}	{0.314, 0.769, 0.146}	{0.152, **0.885**, 0.354}
15	Flicker with swell	{0.925, 0.968, 0.83}	{0.63, 0.625, 0.769}	{0.739, **0.983**, 0.562}	{0.825, 0.98, 0.83}
	Total Score	{0.771, 0.776, 0.584}	{0.71, 0.708, 0.539}	{0.695, **0.826**, 0.584}	{0.693, 0.816, 0.547}

TABLE VIII
AVERAGE SOFT EXPLAINABILITY SCORE PER CNN/XAI/DISTURBANCE FOR THE VALIDATION SET

#	Disturbance	CNN-1	CNN-2	CNN-3	CNN-4
	XAI Techniques		{Occlusion-sensitivity / Grad-CAM / LIME}		
1	Sag	{0.322, 0.24, 0.048}	{0.179, 0.245, 0.008}	{**0.416**, 0.095, 0.1}	{0.299, 0.225, 0.008}
2	Swell	{**0.36**, 0.26, 0.19}	{0.344, 0.315, 0.151}	{0.291, 0.257, 0.111}	{0.357, 0.268, 0.135}
3	Interruption	{0.201, 0.064, 0.001}	{0.152, 0.156, 0.014}	{0.022, **0.323**, 0.002}	{0.27, 0.314, 0.008}
4	Harmonics	{0.62, 0.46, 0.52}	{0.523, 0.543, 0.254}	{**0.736**, 0.481, 0.418}	{0.73, 0.5, 0.23}
5	Flicker	{0.2, 0.29, 0.124}	{0.278, 0.266, 0.05}	{0.29, 0.19, **0.36**}	{0.284, 0.42, 0.144}
6	Oscillatory transient	{**0.41**, 0.08, 0.062}	{0.179, 0.126, 0.0973}	{0.151, 0.1876, 0.104}	{0.175, 0.152, 0.07}
7	Impulsive transient	{0.01, 0.005, 0.015}	{0.0126, 0.01, 0.018}	{**0.02**, 0.006, 0.0153}	{0.015, 0.013, 0.01}
8	Notch (periodic)	{0.047, 0.078, 0.0142}	{0.051, 0.044, 0.0041}	{0.056, **0.144**, 0.045}	{0.042, 0, 0.005}
9	Spike	{0.057, 0.04, 0.053}	{0.05, 0.041, 0.03}	{0.046, **0.113**, 0.08}	{0.043, 0.165, 0.063}
10	Sag with harmonics	{0.311, 0.253, 0.11}	{0.211, 0.163, 0.051}	{ **0.338**, 0.239, 0.036}	{0.21, 0.149, 0.048}
11	Swell with harmonics	{**0.38**, 0.136, 0.105}	{0.31, 0.25, 0.13}	{0.366, 0.124, 0.166}	{0.324, 0.222, 0.135}
12	Interruption with harmonics	{0.16, 0.009, 0.0013}	{**0.172**, 0.21, 0.0055}	{0.145, 0.226, 0.004}	{0.028, 0.27, 0.001}
13	Flicker with harmonics	{0.64, 0.28, 0.47}	{0.173, 0.472, 0.115}	{**0.74**, 0.534, 0.463}	{0.685, 0.5540.115}
14	Flicker with sag	{0.149, **0.15491**, 0.15486}	{0.063, 0.118, 0.009}	{0.044, 0.126, 0.0145}	{0.015, 0.091, 0.024}
15	Flicker with swell	{0.233, **0.34**, 0.155}	{0.117, 0.1, 0.099}	{0.113, 0.268, 0.064}	{0.165, 0.153, 0.123}
	Total Score	{**0.273**, 0.178, 0.134}	{0.19, 0.204, 0.07}	{0.251, 0.221, 0.13}	{0.242, 0.233, 0.074}

TABLE IX
LUT OF THE BEST HARD EXPLANATION COMBINATION PER DISTURBANCE AND EXPLAINABILITY SCORE RESULTS FOR THE TESTING SET

#	Disturbance	Combination	Score
1	Sag	CNN-4 & Grad-CAM	0.98
2	Swell	CNN-3 & Grad-CAM	0.99
3	Interruption	CNN-3 & Grad-CAM	0.944
4	Harmonics	All combinations	1
5	Flicker	All combinations	1
6	Oscillatory transient	CNN-4 & Grad-CAM	0.994
7	Impulsive transient	CNN-2 & LIME	0.025
8	Notch	CNN-3 & Occlusion-sensitivity	0.954
9	Spike	CNN-1 & Occlusion-sensitivity	0.94
10	Sag + Harmonics	CNN-4 & Grad-CAM	0.965
11	Swell + Harmonics	CNN-4 & Grad-CAM	0.98
12	Interruption + Harmonics	CNN-4 & Grad-CAM	0.946
13	Flicker + Harmonics	All combinations	1
14	Flicker + Sag	CNN-4 & Grad-CAM	0.896
15	Flicker + Swell	CNN-3 & Grad-CAM	0.981
		Total Score	**0.906**

TABLE X
LUT OF THE BEST SOFT EXPLANATION COMBINATION PER DISTURBANCE AND EXPLAINABILITY SCORE RESULTS FOR THE TESTING SET

#	Disturbance	Combination	Score
1	Sag	CNN-3 & Occlusion-sensitivity	0.413
2	Swell	CNN-1 & Occlusion-sensitivity	0.32
3	Interruption	CNN-3 & Grad-CAM	0.325
4	Harmonics	CNN-3 & Occlusion-sensitivity	0.74
5	Flicker	CNN-3 & LIME	0.363
6	Oscillatory transient	CNN-1 & Occlusion-sensitivity	0.392
7	Impulsive transient	CNN-3 & Occlusion-sensitivity	0.0126
8	Notch	CNN-3 & Grad-CAM	0.133
9	Spike	CNN-3 & Grad-CAM	0.112
10	Sag + Harmonics	CNN-3 & Occlusion-Sensitivity	0.321
11	Swell + Harmonics	CNN-1 & Occlusion-sensitivity	0.36
12	Interruption + Harmonics	CNN-2 & Occlusion-sensitivity	0.174
13	Flicker + Harmonics	CNN-3 & Occlusion-sensitivity	0.74
14	Flicker + Sag	CNN-1 & Grad-CAM	0.145
15	Flicker + Swell	CNN-1 & Grad-CAM	0.34
		Total Score	**0.3253**

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
EXPLAINABILITY SCORE PER CNN/XAI FOR REAL LIFE ELECTRICAL SAGS [35]

	CNN-1	CNN-2	CNN-3	CNN-4
XAI Techniques	{Occlusion-sensitivity / Grad-CAM / LIME}			
Hard explainability score (Binary)	$\{0.289, 0.911, 0.467\}$	$\{0.689, 0.778, 0.711\}$	$\{0.711, 0.711, 0.467\}$	$\{0.578, \mathbf{0.978}, 0.711\}$
Soft explainability score (IoU)	$\{0.063, 0.19, 0.038\}$	$\{0.183, 0.231, 0.116\}$	$\{\mathbf{0.334}, 0.239, 0.059\}$	$\{0.282, 0.262, 0.064\}$

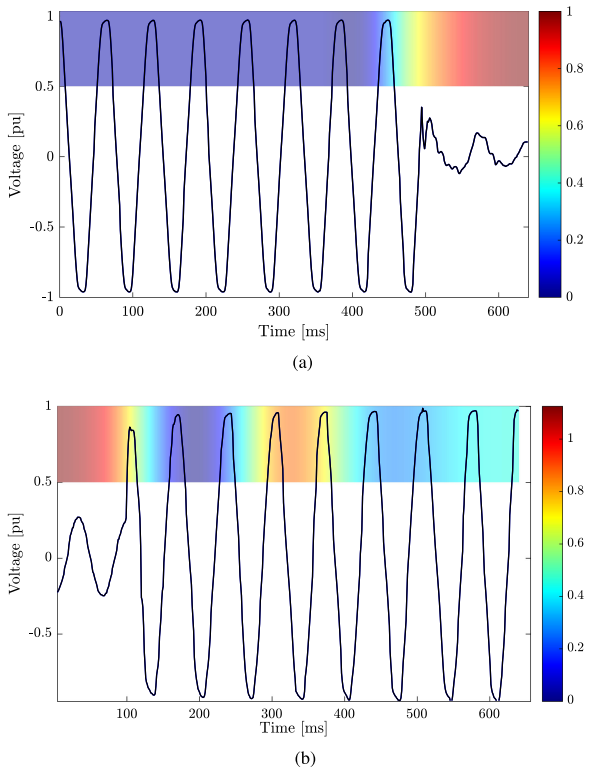


Fig. 6. Explanation of predictions for real life Sag signals: Examples. (a) CNN-3 with Occlusion-sensitivity. The soft explainability score is 1. (b) CNN-2 with Grad-CAM. The soft explainability score is 0.62.

Real-life sag events, one the most common PQD, have been examined on the suggested method using the data from [35]. This dataset provides signals recordings from the power network of the University of Cádiz during five years and includes a diverse set of real sag events in a network of 50 Hz. An example of real sag which is classified using the trained CNN-3 and explained by occlusion-sensitivity is presented in Fig. 6(a). In this example, the main features that explain the prediction are colored in red and the soft explainability *score* is 1, a perfect explanation. Another example is presented in Fig. 6(b). In this example, a real sag signal is explained by Grad-CAM after using CNN2. In this example, the soft explainability *score* is 0.62 since some of the strongest features are not related directly to the disturbance. For the experiment, a collection of 45 sags events were used and the average score for each classifier and each XAI technique was calculated based on Algorithm 3, where $D = 1$. The rows in Table XI present the hard and soft explainability *scores* accordingly. It can be seen that the combinations with the highest score for both metrics using real data are identical to the combinations with the highest score using the sag signals from the open source generator. That is to say, for both datasets, the combinations with the highest hard and soft explainability *score* for sag are CNN4 & Grad-Cam and CNN3 & occlusion-sensitivity, respectively. The hard explainability *score*s are 0.987, 0.978 and the soft explainability *score*s are 0.416, 0.334 for the generator and real sag signals, accordingly.

TABLE XII
PROPOSED HIGHWAY NETWORK ARCHITECTURE

#	Layer	Parameters	Activation
1	Fully connected	size = 256	ReLU, dropout = 0.25
2	Highway layer (size = 256)		
3	Fully connected	size = 128	ReLU, dropout = 0.25
4	Highway layer (size = 128)		
5	Fully connected	size = 64	ReLU, dropout = 0.25
6	Highway layer (size = 64)		
7	Fully connected	size = 32	ReLU
8	Highway layer (size = 32)		
9	Fully connected	Size = 16	Softmax

D. Comparison With Other Deep Neural Network Methods

CNN models are considered as the state-of-the-art for classification of PQDs, thus this type of neural network was proposed in this article. Nonetheless, there are other deep neural networks, which can serve for the proposed XAI method. In this section, we will calculate the explainability score for a neural network model which is based on the highway network concept. Inspired by long short-term memory and recurrent neural networks, a highway network consists of layers that increase the network depth and optimize it by using a mechanism to regulate information flow [36]. This mechanism is based on the structure of a hidden layer with a layer's input x and layer's output y which follows [36, (2)]:

$$y = H(x, W_H)T(x, W_T) + xC(x, W_C) \tag{12}$$

where H, C, T apply nonlinear transforms and W_H, W_T, W_C are trained weights. T is referred as the transform gate and C as the carry gate, since they express how much of the output is produced by transforming the input and carrying it. This structure can solve or at least partially prevent the vanishing gradient problem, thus leading to easy optimization of neural networks.

The architecture of the developed highway network is presented in Table XII. In this network, the following activation functions were chosen: H is tanh, T is sigmoid, C is the identity function. Fully connected layers are used for the weights calculation. The parameters in the training process are as follows:

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

TABLE XIII
TOTAL EXPLAINABILITY SCORE USING THE HIGHWAY NETWORK

Dataset/ XAI	Occlusion-sensitivity	LIME
Average explainability score {hard / soft}		
Validation Set	$\{0.687, 0.214\}$	$\{0.536, 0.127\}$
Real life sags	$\{0.4, 0.119\}$	$\{0.711, 0.075\}$

number of epochs is 50, mini batch size is 128, $m_0 = 0.001$, $df = 0.1$, and $dp = 20$. The average accuracy of this network achieves more than 99.43% for the training set and accuracy around 96.1% for the validation and testing set. After training the network, the total score for each XAI technique was calculated for the validation set and the real life sag events based on (11) and shown in Table XIII. Note that some XAI techniques such as Grad-CAM are dedicated specifically to CNN models and can not be used on this classifier. Based on the total scores results, it can be seen that most of the CNNs outperformed the suggested highway network by means of explainability score. For example, the hard explainability score achieves a lower score than any CNN network for the validation set. Furthermore, there is no combination of XAI technique with the highway network that achieves the highest explainability score and therefore the highway network will not be used in the inference stage. Following this experiment, CNN models can be considered as more trustworthy and transparent when compared to a highway network with fully connected layers.

V. CONCLUSION

Advanced ML techniques have recently demonstrated outstanding performance when applied to PQD classification. Nevertheless, power experts may find it hard to trust the results of such algorithms, if they do not fully understand the reasons for a certain algorithm's output. In this light, this article suggests a method that explains the outputs of CNN based classifiers, using XAI. A key result is that the suggested XAI method retains high classification accuracy, while its outputs are optimized to be easy to understand. We show that based on the "hard explainability" score, the Grad-Cam XAI technique is the most suitable for classification of PQD, while based on the "soft explainability" score, the occlusion-sensitivity XAI method is the most suitable for this application. By using the suggested evaluation metrics, the appropriate classifier and XAI technique for each disturbance can be chosen such that the the explainability score improves significantly. Furthermore, we showed that the total explainability score of the proposed method was better when compared to any other combination of XAI technique and CNN classifier. Preserving the classification accuracy and improving the transparency of the classifier using the proposed XAI method will allow power experts to make informed and trustworthy decisions in cases, where a PQD event occurs in the grid. The presented XAI method can be implemented to various applications in the field of power systems with a focus on fault and abnormal event classification tasks. The main limitation of the method is the requirement of multiple combinations of classifiers and XAI techniques. We believe that XAI techniques have significant potential to explain the decisions of ML algorithms, which are increasingly being used nowadays in the power system community. Therefore, there may be significant room for future research, which may focus on the use of XAI techniques in other energy related applications.

REFERENCES

[1] M. Khodayar, G. Liu, J. Wang, and M. E. Khodayar, "Deep learning in power systems research: A review," *CSEE J. Power Energy Syst.*, vol. 7, no. 2, pp. 209–220, 2020.

[2] A. K. Ozcanli, F. Yaprakdal, and M. Baysal, "Deep learning methods and applications for electrical power systems: A comprehensive review," *Int. J. Energ. Res.*, vol. 44, no. 9, pp. 7136–7157, 2020.

[3] D. Gunning, "Explainable artificial intelligence (XAI)" Defense Advanced Research Projects Agency, Tech. Rep. DARPA-BAA-16-53, 2017. [Online]. Available: https://www.darpa.mil/program/explainable-artificial-intelligence

[4] A. Adadi and M. Berrada, "Peeking inside the black-box: A survey on explainable artificial intelligence (XAI)," *IEEE Access*, vol. 6, pp. 52138–52160, 2018, doi: 10.1109/ACCESS.2018.2870052.

[5] A. Das and P. Rad, "Opportunities and challenges in explainable artificial intelligence (XAI): A survey," 2020, *arXiv:2006.11371*.

[6] Y. G. Akhlaghi *et al.*, "Hourly performance forecast of a dew point cooler using explainable artificial intelligence and evolutionary optimisations by 2050," *Appl. Energ.*, vol. 281, 2021, Art. no. 116062.

[7] M. Kuzlu, U. Cali, V. Sharma, and O. Guler, "Gaining insight into solar photovoltaic power generation forecasting utilizing explainable artificial intelligence tools," *IEEE Access*, vol. 8, pp. 187814–187823, 2020, doi: 10.1109/ACCESS.2020.3031477.

[8] N. Mohan, K. P. Soman, and R. Vinayakumar, "Deep power: Deep learning architectures for power quality disturbances classification," in *Proc. Int. Conf. Technol. Adv. Power Energy*, 2017, pp. 1–6.

[9] Y. Deng, L. Wang, H. Jia, X. Tong, and F. Li, "A sequence-to-sequence deep learning architecture based on bidirectional GRU for type recognition and time location of combined power quality disturbance," *IEEE Trans. Ind. Informat.*, vol. 15, no. 8, pp. 4481–4493, Aug. 2019.

[10] Y. Dong, H. Cao, X. Ding, G. Xu, and C. Yue, "Multi-task learning method for classification of multiple power quality disturbances," *IET Gener. Transmiss. Distrib.*, vol. 14, no. 5, pp. 900–909, 2020.

[11] W. Qiu, Q. Tang, J. Liu, and W. Yao, "An automatic identification framework for complex power quality disturbances based on multifusion convolutional neural network," *IEEE Trans. Ind. Informat.*, vol. 16, no. 5, pp. 3233–3241, May 2020.

[12] R. Gong and T. Ruan, "A new convolutional network structure for power quality disturbance identification and classification in microgrids," *IEEE Access*, vol. 8, pp. 88801–88814, 2020, doi: 10.1109/ACCESS.2020.2993202.

[13] H. Li, B. Yi, Q. Li, J. Ming, and Z. Zhao, "Evaluation of DC power quality based on empirical mode decomposition and one-dimensional convolutional neural network," *IEEE Access*, vol. 8, pp. 34339–34349, 2020, doi: 10.1109/ACCESS.2020.2974571.

[14] H. Liu, F. Hussain, Y. Shen, S. Arif, A. Nazir, and M. Abubakar, "Complex power quality disturbances classification via curvelet transform and deep learning," *Electr. Power Syst. Res.*, vol. 163, pp. 1–9, 2018.

[15] J. Wang, Z. Xu, and Y. Che, "Power quality disturbance classification based on compressed sensing and deep convolution neural networks," *IEEE Access*, vol. 7, pp. 78336–78346, 2019, doi: 10.1109/ACCESS.2019.2922367.

[16] K. Cai, W. Cao, L. Aarniovuori, H. Pang, Y. Lin, and G. Li, "Classification of power quality disturbances using Wigner-Ville distribution and deep convolutional neural networks," *IEEE Access*, vol. 7, pp. 119099–119109, 2019, doi: 10.1109/ACCESS.2019.2937193.

[17] S. Wang and H. Chen, "A novel deep learning method for the classification of power quality disturbances using deep convolutional neural network," *Appl. Energ.*, vol. 235, pp. 1126–1140, 2019.

[18] X.-H. Li *et al.*, "A survey of data-driven and knowledge-aware eXplainable AI," *IEEE Trans. Knowl. Data Eng.*, to be published, doi: 10.1109/TKDE.2020.2983930.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

[19] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Computer Vision*. Berlin, Germany: Springer, 2014, pp. 818–833.
[20] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?": Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 1135–1144.
[21] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 618–626.
[22] M. H. J. Bollen, *Understanding Power Quality Problems*. Hoboken, NJ, USA: Wiley, 1999.
[23] M. H. J. Bollen and I. Y.-H. Gu, *Signal Processing of Power Quality Disturbances*. Hoboken, NJ, USA: Wiley, 2006.
[24] E. Hossain, M. R. Tur, S. Padmanaban, S. Ay, and I. Khan, "Analysis and mitigation of power quality issues in distributed generation systems using custom power devices," *IEEE Access*, vol. 6, pp. 16816–16833, 2018, doi: 10.1109/ACCESS.2018.2814981.
[25] F. A. S. Borges, R. A. S. Fernandes, I. N. Silva, and C. B. S. Silva, "Feature extraction and power quality disturbances classification using smart meters signals," *IEEE Trans. Ind. Informat.*, vol. 12, no. 2, pp. 824–833, Apr. 2016.
[26] S. Khokhar, A. A. B. M. Zin, A. S. B. Mokhtar, and M. Pesaran, "A comprehensive overview on signal processing and artificial intelligence techniques applications in classification of power quality disturbances," *Renewable Sustain. Energy Rev.*, vol. 51, pp. 1650–1663, 2015.
[27] R. Igual and C. Medrano, "Research challenges in real-time classification of power quality disturbances applicable to microgrids: A systematic review," *Renewable Sustain. Energy Rev.*, vol. 132, 2020, Art. no. 110050.
[28] G. S. Chawda *et al.*, "Comprehensive review on detection and classification of power quality disturbances in utility grid with renewable energy penetration," *IEEE Access*, vol. 8, pp. 146807–146830, 2020, doi: 10.1109/ACCESS.2020.3014732.
[29] R. Machlev, A. Chachkes, J. Belikov, Y. Beck, and Y. Levron, "Open source dataset generator for power quality disturbances with deep-learning reference classifiers," *Electr. Power Syst. Res.*, vol. 195, 2021, Art. no. 1057152.
[30] *IEEE Recommended Practice for Monitoring Electric Power Quality*, IEEE Standard 1159-1995, 2009.
[31] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 658–666.
[32] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[33] C. M. Bishop, *Neural Networks for Pattern Recognition*. London, U.K.: Oxford Univ. Press, 1995.
[34] G. Hripcsak and A. S. Rothschild, "Agreement, the f-measure, and reliability in information retrieval," *J. Am. Med. Informat. Assoc.*, vol. 12, no. 3, pp. 296–298, 2005.
[35] O. Florencias-Oliveros, M. J. Espinosa-Gavira, J.-J. G. de la Rosa, A. Agüera-Pérez, J. C. Palomares-Salas, and J. M. Sierra-Fernández, "Real-life power quality Sags," IEEE Dataport, 2017. [Online]. Available: https://dx.doi.org/10.21227/H2K88D
[36] R.K. Srivastava, K. Greff, and J. Schmidhuber, "Highway networks," 2015, *arXiv:1505.00387*.



Ram Machlev (Student Member, IEEE) received the B.S. and M.S. (*cum laude*) degrees in electrical engineering from Tel-Aviv University, Tel-Aviv, Israel, in 2013 and 2018, respectively. He is currently working toward the Ph.D. degree in explainable artificial intelligence for power and energy monitoring systems with the Faculty of Electrical and Computer Engineering, Technion—Israel Institute of Technology, Haifa, Israel.

In the years 2012–2020, he was an Engineer with Motorola, Broadcom and Sony. His research interests include smart grids and power systems.



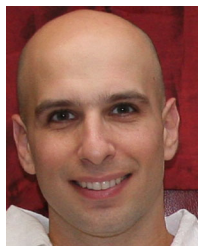
Michael Perl received the B.S. degree (*cum laude*) in electrical engineering and physics in 2020 from the Technion–Israel Institute of Technology, Haifa, Israel, where he is currently working toward the M.S. degree in electrical and computer engineering.

His current research interests include power systems and machine learning.



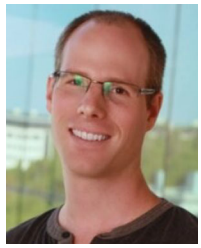
Juri Belikov (Senior Member, IEEE) received the B.S. degree in mathematics from Tallinn University, Tallinn, Estonia, in 2006, and the M.S. and Ph.D. degrees in computer and systems engineering from the Tallinn University of Technology, Tallinn, in 2008 and 2012, respectively.

In the years 2015–2017, he was a Postdoctoral Fellow with the Faculty of Mechanical Engineering and Andrew & Erna Viterbi Faculty of Electrical Engineering, Technion—Israel Institute of Technology, Haifa, Israel. He is currently an Assistant Professor with the Department of Software Science, Tallinn University of Technology. His main research interests include edge of nonlinear control theory and power systems.



Kfir Yehuda Levy received the B.S. and M.S. degrees in electrical engineering from Technion–Israel Institute of Technology, Haifa, Israel, in 2003 and 2011 respectively, and the Ph.D. degree in industrial engineering and management from Technion–Israel Institute of Technology, Haifa, Israel, in 2016.

From 2016 to 2019, he was a Postdoctoral Fellow with ETH Zurich, Zurich, Switzerland. He is currently an Assistant Professor with the Faculty of Electrical and Computer Engineering, Technion–Israel Institute of Technology. His research interests include machine learning and optimization.



Yoash Levron (Senior Member, IEEE) received the B.S. degree in electrical engineering from Technion, Israel Institute of Technology, Haifa, Israel, in 2001, and the M.S. and Ph.D. degrees in engineering from Tel-Aviv University, Tel Aviv, Israel, in 2007 and 2013, respectively.

In the years 2013–2014, he was a Postdoctoral Fellow with the University of Colorado at Boulder, CO, USA. He is currently an Associate Professor with the Faculty of Electrical and Computer Engineering, Technion, Israel Institute of Technology. His research interests include power systems, power electronics, and energy conversion.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.